Application Flow & User Journey Specification

Personal Health Record Management and Sharing System (PHRMS)

Course: MIS 4500 – Technology Capstone Project (Fall 2026)

Author: Abishekan Sri Baskaran (Student ID: 0001044)

Supervisor: Dr. Ama Kulathilake

1. Master Whole-Application Flowchart

Figure 1 illustrates the complete end-to-end architecture and workflow of the PHRMS web application across all three stakeholder roles (**Patient**, **Document Review Team**, and **Doctor**), from role-based login through document review, chronological record organization, consultation summary generation, doctor access control, and activity logging.

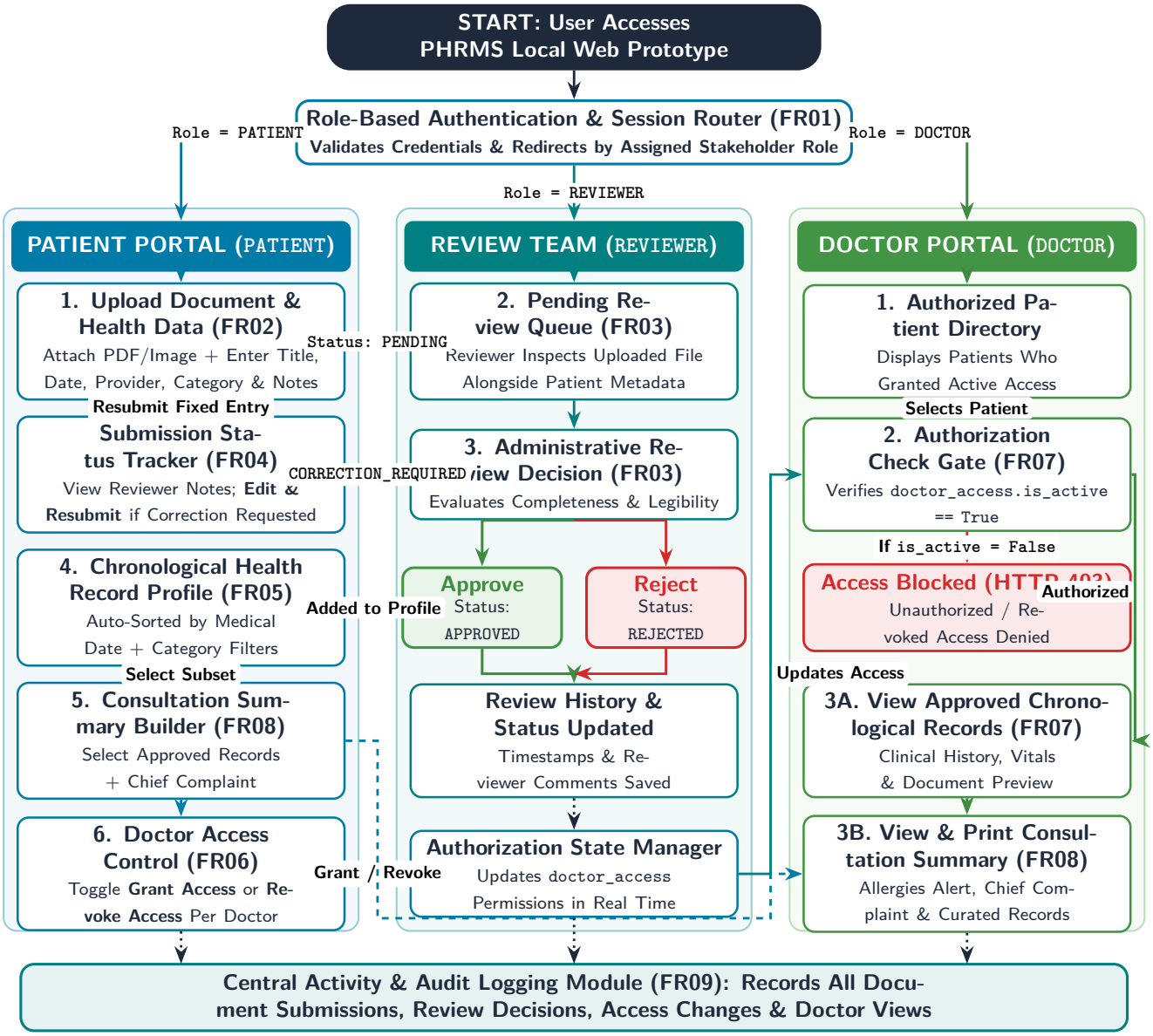


Figure 1: Complete End-to-End Application Flowchart across Patient, Document Review Team, and Doctor Portals.

2. Document Review State Lifecycle Flowchart

Figure 2 details the state transition logic for medical documents submitted by a patient, illustrating how administrative review isolates unverified or incomplete uploads from the official chronological patient record.

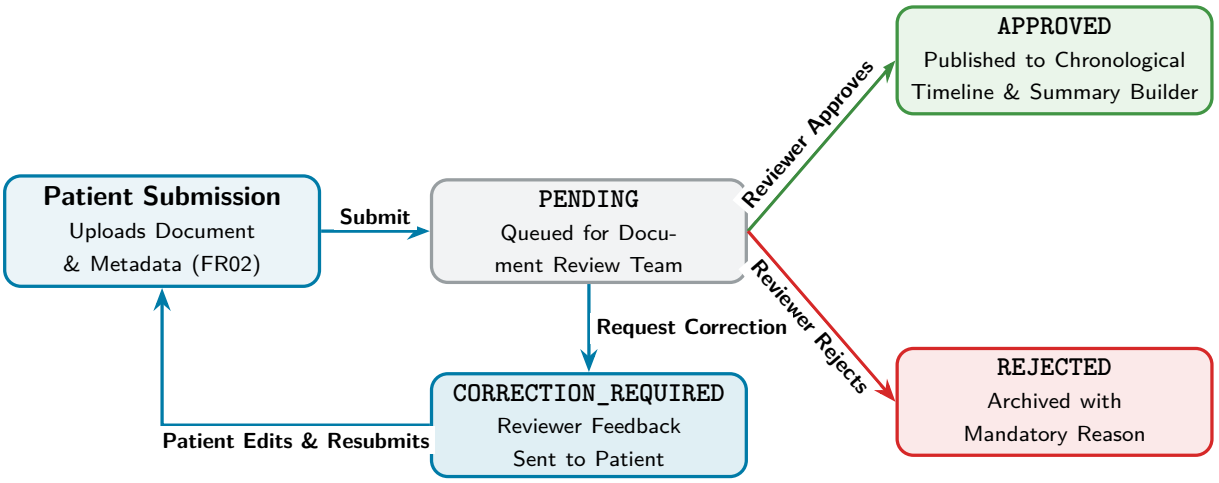


Figure 2: Health Record Administrative Review State Machine (`PENDING` → `APPROVED` / `CORRECTION_REQUIRED` / `REJECTED`).

3. High-Level System Workflow Summary

The PHRMS prototype connects three distinct user portals through a unified state machine for health records and an explicit authorization gate for physician access.

Stage / Actor	Transition / Action	System State & Outcome
1. Patient Submission	Uploads Document & Metadata	Record enters **PENDING** status and appears in the Document Review Team queue.
2A. Reviewer: Approve	Selects **Approve**	Record transitions to **APPROVED**; automatically added to Chronological Patient Timeline and eligible for Consultation Summary Builder.
2B. Reviewer: Correction	Selects **Request Correction**	Record transitions to **CORRECTION_REQUIRED** with reviewer instructions. Patient edits & resubmits → returns to **PENDING**.
2C. Reviewer: Reject	Selects **Reject**	Record transitions to **REJECTED** and is archived with mandatory reviewer reason.
3A. Access Control: Grant	Patient toggles **Grant Access**	Authorized **Doctor** can view the patient's Approved Chronological Timeline and Consultation Summaries (every view is logged).
3B. Access Control: Revoke	Patient toggles **Revoke Access**	Doctor's access is immediately terminated and the patient is removed from the Doctor's active directory.

4. Detailed Role-by-Role Screen Flows

4.1 Authentication & Role Routing Flow (FR01)

1. **Landing / Login Screen:** User enters Email & Password (or clicks one of the **Quick Demo Role Switcher** cards for *Patient*, *Document Review Team*, or *Doctor* during academic demonstration).
2. **Role Router:**
   - If `role == 'PATIENT'` → Redirect to **Patient Portal Dashboard**.
   - If `role == 'REVIEWER'` → Redirect to **Document Review Team Workspace**.
   - If `role == 'DOCTOR'` → Redirect to **Physician Consultation Portal**.

4.2 Patient Portal Flow (`PATIENT`)

The Patient Portal is organized into five clear navigation views:

1. **Overview & Submission Status Tracker (FR04):**

- Displays summary cards: *Total Approved Records*, *Pending Administrative Review*, *Needs Correction*, and *Authorized Doctors*.
- **Submission Tracker Table:** Shows all submitted records with color-coded status badges (`Approved`, `Pending Review`, `Correction Requested`, `Rejected`) and any notes left by the Document Review Team.
- For items marked `Correction Requested`, an inline **"Edit & Resubmit"** modal allows the patient to fix metadata or re-upload a clearer document, transitioning the item back to `Pending Review`.

2. **Upload Medical Document / Add Health Entry (FR02):**
   - Form inputs: *Document Title*, *Category* (Lab Report, Prescription, Consultation Note, Imaging / Scan, Vaccination / Vitals), *Healthcare Provider / Lab Name*, *Date of Medical Record*, *Clinical Summary / Key Metrics*, and *Document File Attachment*.
   - Submitting creates a `PENDING` record and logs `RECORD_SUBMITTED`.
3. **Chronological Health Timeline (FR05):**
   - Displays only `APPROVED` records sorted in reverse chronological order ($t_{\text{newest}} \rightarrow t_{\text{oldest}}$) by the medical event date.
   - Includes category filter pills (*All*, *Lab Reports*, *Prescriptions*, *Consultation Notes*, *Imaging*, *Vitals*) and keyword search.
   - Each timeline card shows the date, category badge, provider, clinical summary, reviewer verification badge, and a button to view the attached document.
4. **Consultation-Ready Summary Builder (FR08):**
   - **Step 1:** Patient enters a *Summary Title* (e.g., "Cardiology Follow-Up Visit") and *Chief Complaint / Questions for Doctor*.
   - **Step 2:** Patient checks the boxes next to specific `APPROVED` records from their chronological history that are relevant to this appointment.
   - **Step 3:** Clicking **"Generate Consultation Summary"** compiles a structured, print-ready clinical summary card containing Patient Demographics, Blood Group, Allergies, Chief Complaint, and the selected chronological records.
5. **Doctor Access Control & Audit Log (FR06, FR09):**
   - **Doctor Authorization Panel:** Lists registered doctors with their specialization and current status (`Authorized` in Health Green vs. `No Access` in Cool Gray). Clicking **Grant Access** or **Revoke Access** immediately updates permissions.
   - **Transparency Audit Log:** Displays a chronological table of all actions taken on the patient's profile (who reviewed which document, when a doctor viewed the timeline, and when access was changed).

4.3 Document Review Team Flow (`REVIEWER`)

The Review Team interface focuses on administrative quality control (FR03):

1. **Pending Review Queue:** Displays a prioritized queue of all patient submissions currently in `PENDING` status.
2. **Document Inspection Drawer / Modal:**

- Shows side-by-side comparison of **Patient-Entered Metadata** (Patient Name, Record Title, Category, Provider, Record Date, Clinical Summary) and **Uploaded Document Preview**.
- **Administrative Checklist:** Legibility check, date match, category match, and completeness check.

3. **Review Decision Actions:**
   - **Approve (`#4CAF50` Health Green):** Marks record `APPROVED`, stamps reviewer ID and timestamp, and publishes it to the patient's chronological timeline.
   - **Request Correction (`#007BA7` Surgical Blue):** Marks record `CORRECTION_REQUIRED` and requires the reviewer to specify what the patient needs to fix.
   - **Reject (`#D62828` Aid Red):** Marks record `REJECTED` with a mandatory reason.
4. **Completed Reviews Log:** Shows historical decisions made by the review team for auditing.

4.4 Doctor Portal Flow (`DOCTOR`)

The Physician interface provides fast, distraction-free clinical preparation (FR07, FR08):

1. **Authorized Patients Directory:**
   - Displays cards for patients who have actively granted access to the logged-in doctor.
   - If a patient revokes access, their card immediately disappears from the doctor's active directory.
2. **Patient Clinical Workspace (Authorized View):**
   - Selecting an authorized patient automatically records a `DOCTOR_VIEWED_TIMELINE` audit event.
   - **Patient Header Banner:** Displays Patient Name, Age/DOB, Blood Type, and a prominent **Known Allergies** alert badge.
   - **Tab A – Consultation-Ready Summaries (FR08):** Highlights pre-consultation summaries prepared by the patient so the doctor can review the chief complaint and curated records in under 60 seconds, including a **Print / Export View** button.
   - **Tab B – Full Chronological Health Record (FR05):** Displays the complete timeline of `APPROVED` records with category filtering and document preview. (Unapproved, pending, or rejected records are never visible to the doctor).